



Packing two copies of a tree into a bipartite graph with restrained maximum degree[☆]

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ABSTRACT

For a bipartite graph $G(U_1, U_2)$, we use $\Delta_1(G)$ and $\Delta_2(G)$ to denote the maximal degree of the vertices in the sets U_1 and U_2 in G , respectively. Let the tree $T(V_1, V_2)$ be an (a, b) -bipartite graph of order n . We prove that if $|a - b| \leq 2$, then there is a 2-packing (σ, τ) of T in some complete bipartite graph $B_{n+1}(X_1, X_2)$ such that for each $i \in \{1, 2\}$, $\Delta_i(\sigma(T) \cup \tau(T))$ is at most $\Delta_i(T) + 2$, where $V_i \subseteq X_i$ ($i = 1, 2$). This result is sharp because $\Delta_i(T) + 2$ cannot be reduced any further. By applying this result, we deduce a corollary for the existence of packing of three bipartite graphs into complete bipartite graphs.

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1. Introduction

We consider only simple finite graphs and use standard terminology and notation from [1] except for what is explicitly stated. Let $G = (V(G), E(G))$ be a graph, we denote the order and size of G by $|V(G)|$ and $e(G)$, respectively. We use $G(X, Y)$ to denote a bipartite graph with bipartition (X, Y) . If (X', Y') is also a bipartition of G but $(X, Y) \neq (X', Y')$, then we regard $G(X, Y) \neq G(X', Y')$. Moreover, if $|X| = a$ and $|Y| = b$, then we say that G admits an (a, b) -bipartition and call G is an (a, b) -bipartite graph. We use $B_n(X, Y)$ to denote the complete bipartite with bipartition (X, Y) and order n . Note that up to isomorphism, B_n is not uniquely defined for $n \geq 4$. A forest is a graph without cycles. A tree is a connected forest. Obviously, a tree is also a bipartite graph.

Let $G(U_1, U_2)$ be a bipartite graph. For $i \in \{1, 2\}$, we denote by $\Delta_i(G)$ and $\delta_i(G)$ the maximum and minimum degrees of the vertices in U_i in G , respectively. In addition, we give the following definitions.

Definition 1 ([9]). We say that σ is an embedding of G in $B_n(X_1, X_2)$ means that σ is a mapping from $V(G)$ into $V(B_n)$ such that $\sigma(U_i) \subseteq X_i$ ($i = 1, 2$). Under this mapping we define a subgraph $\sigma(G)$ of B_n , where $\sigma(V(G))$ as its vertex set and $\{\sigma(u)\sigma(v) \mid uv \in E(G)\}$ as its edge set. Clearly $\sigma(G)$ is isomorphic to G .

Definition 2 ([11]). Let $H(V_1, V_2)$ be a bipartite graph. We say that there is a packing (σ, τ) of G and H in B_n if there exist embeddings σ and τ of G and H in B_n such that $\sigma(G)$ and $\tau(H)$ are edge-disjoint, respectively. When $G = H$, we say that there is a 2-packing of G in B_n and (σ, τ) is the 2-packing of G in B_n .

Definition 3. By packing H into G we mean that there exists a packing (σ, τ) of G and H in $B_{|G|}(U_1, U_2)$ where σ is the identity embedding.

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Table 1
A 2-packing (σ_1, σ_2) of $G(A, B)$ in $B_n(X, Y)$.

	a_1	$\cdots$	a_s	b_1	$\cdots$	b_q
σ_1	x_i	$\cdots$	$*$	y_j	$\cdots$	$*$
σ_2	$*$	$\cdots$	$*$	$*$	$\cdots$	$*$

Definition 4. Let $G(A, B)$ is a (s, t) -bipartite graph with $A = \{a_1, \dots, a_s\}$ and $B = \{b_1, \dots, b_t\}$ and let $B_n(X, Y)$ be a complete bipartite graph of order n with $X = \{x_1, \dots, x_p\}$ and $Y = \{y_1, \dots, y_q\}$, where $s \leq p$ and $t \leq q$. Then the 2-packing (σ_1, σ_2) of G in B_n can be represented as follows by using two matrices (see Table 1).

Let $L = (l_{ij})_{2 \times s}$ ($l_{ij} \in X$) and $R = (r_{ij})_{2 \times t}$ ($r_{ij} \in Y$) be the matrices on the left and right of the above figure, respectively, then $l_{ij} = \sigma_1(a_j)$ and $r_{ij} = \sigma_2(b_j)$.

Note that for the matrices L, R we still have a hard time checking whether $\sigma_1(G)$ and $\sigma_2(G)$ are edge-disjoint, and for this reason we will also represent the 2-packing of G with some pictures. If in the picture $\sigma_1(G)$ and $\sigma_2(G)$ are two edge disjoint graphs, then it can be shown that (σ_1, σ_2) is a 2-packing of G in B_n .

First we give some results on the packing of bipartite graphs presented by Fouquet and Wojsa in [4]. Before introducing their theorem, we need to introduce the special graph in [4], which we define as the set of graphs Ω . If $G \in \Omega$, then G is isomorphic to one of the graphs F_i ($1 \leq i \leq 4$) constructed below.

- We use the (a, b) -bipartite graph F_1 to denote the union of two vertex-disjoint stars whose center vertices are not in the same part, where $a \geq 3$ and $b \geq 3$.
- Let r, s , and t be three non-negative integers such that $r + s + t \geq 1$. We denote by $F_2(X, Y)$ a $(3, 2(r + s + t) - 1)$ -bipartite graph of size $2(r + s + t)$ such that $X = \{a_1, a_2, a_3\}$, $Y = A_1 \cup A_2 \cup A_3 \cup B$, $|A_1| = r$, $|A_2| = s$, $|A_3| = t$, $|B| = r + s + t - 1$, $N_{F_1}(a_i) = A_i \cup A_{i+1}$ for $i = 1, 2, 3 \pmod{3}$ and the vertices of B are isolated.
- Let r and s be two non-negative integers, where $r \geq 1$. Let $F_3(X, Y)$ is the $(3, 2r + s - 1)$ -bipartite graph of size $2r + s$ such that $X = \{a, b, c\}$, $Y = A \cup B \cup C$, $|A| = r$, $|B| = s$, $|C| = r - 1$, and xy with $x \in X$, $y \in Y$ is an edge of F_3 if and only if $x \in \{a, b\}$ and $y \in A$ or $x = c$ and $y \in B$.
- Let $F_4(X, Y)$ is the $(3, 2r + 2s - 1)$ -bipartite graph of size $2r + 2s$ in which $X = \{a, b, c\}$, while Y is the union of four mutually disjoint sets A, B, C and D with $|A| = |C| = r$, $|B| = |D| + 1 = s$, $N_{F_4}(a) = A \cup B$, $N_{F_4}(b) = B \cup C$, while c is isolated vertex.

For a survey of the definition of the graph set Ω , we refer the reader to [4].

Let Γ be the family of a pair (a, b) -bipartite graphs $\{G_1(X_1, X_2), G_2(Y_1, Y_2)\}$, where G_1 contains a star with center vertex at X_1 and size b and G_2 is a bipartite graph with $\delta_1(G_2) > 0$, or G_1 contains a star with center vertex at X_2 and size a and G_2 is a bipartite graph with $\delta_2(G_2) > 0$.

Theorem 1.1 ([4]). Let $G(U_1, U_2)$ be an (a, b) -bipartite graph with $a, b \geq 2$ and

$$e(G) \leq a + b - 2.$$

Then either there is a 2-packing of $G(X_1, X_2)$ in $B_{a+b}(X_1, X_2)$, or $G \in \Omega$.

Theorem 1.2 ([4]). Let G and H be two (a, b) -bipartite graphs. If

$$e(G) + e(H) \leq a + b,$$

then there is a packing of G and H in B_{a+b} unless $\{G, H\} \in \Gamma$.

Theorem 1.3 ([4]). Let $G(U_1, U_2)$ and $H(V_1, V_2)$ be two (a, b) -bipartite graphs such that

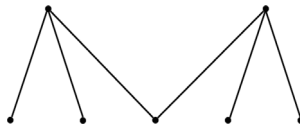
$$\Delta_1(G)\Delta_2(H) + \Delta_1(H)\Delta_2(G) \leq \max(a, b).$$

Then there is a packing of G and H in $B_{a+b}(X_1, X_2)$, where $|X_1| = a$ and $|X_2| = b$.

Another theorem about packing in bipartite graphs was introduced by Wang. A bipartite graph is called a C_4 -free bipartite graph meaning that it does not contain cycles of length 4.

Theorem 1.4 ([10]). Let $G_1(X_1, Y_1)$ and $G_2(X_2, Y_2)$ be two C_4 -free bipartite graphs of order n with $|X_1| = |X_2|$ and $|Y_1| = |Y_2|$. If $e(G_1) + e(G_2) \leq 2n - 2$, then either there is a packing of G_1 and G_2 in some B_{n+1} , or one is the union of vertex-disjoint stars and the other is the union of two vertex-disjoint cycles.

Let $G_1(X_1, Y_1)$ and $G_2(X_2, Y_2)$ be two bipartite graphs whose edges do not intersect. We denote the edge-sum of G_1 and G_2 by the bipartite graph $G_1 \cup G_2$. Let (Z_1, Z_2) as its bipartition, then $Z_1 = X_1 \cup X_2$, $Z_2 = Y_1 \cup Y_2$ and $E(G_1 \cup G_2) = E(G_1) \cup E(G_2)$. Thus for a 2-packing (σ_1, σ_2) of G in B_n , we are allowed to write $\sigma_1(G) \cup \sigma_2(G)$ for the edge-sum of $\sigma_1(G)$ and $\sigma_2(G)$.

Fig. 1. T .

There have been a number of results on the maximum degree in the packing problem, see [2,3,8]. The latest result is that obtained by H. Kaul and H. Reiniger in [6], who proved that if G is a graph and F is a forest, both on n vertices, and if $3\Delta(G) + \ell^*(F) \leq n$ ($\ell^*(F)$ denotes the excess leaves of forest F , i.e., $\ell^*(F) = \sum_{v \in V(F)} \max\{d_F(v) - 2, 0\}$), then there is a packing of G and F in K_n unless n is even, $G = \frac{n}{2}K_2$, and $F = K_{1,n-1}$. Wang proves in [12] that if a tree T is not a star, then there is a packing σ such that $\Delta(T \cup \sigma(T))$ is at most $\Delta(T) + 2$. This result is also obtained by A. Görlich, M. Woźniak in [5]. The inference of the existence of packing of the three graphs can be easily obtained from this result. Inspired by this result, we firstly proved in [7] that for a (a, b) -bipartite graph G with order n , if $b \geq a - 1$, then there is a 2-packing (σ, τ) of G in some B_{n+1} such that $\Delta_2(\sigma(T) \cup \tau(T)) \leq \Delta_2(T) + 2$. Meanwhile, there are many examples to show that in this case we can only restrict the maximum degree of the part with greater order in T , see [7]. However, if $|a - b| \leq 2$, the situation becomes very different and we hope to obtain a 2-packing with restrained maximum degree in both of two parts in T . To be precise, we prove the following theorem.

Theorem 1.5. *Let the tree $T(V_1, V_2)$ be an (a, b) -bipartite graph of order n . If $|a - b| \leq 2$, then there is a 2-packing (σ, τ) of T in some $B_{n+1}(X_1, X_2)$ such that for any $i \in \{1, 2\}$,*

$$\Delta_i(\sigma(T) \cup \tau(T)) \leq \Delta_i(T) + 2,$$

where $V_i \subseteq X_i$ ($i = 1, 2$).

By observing a path of length at least 4 one gets that $\Delta_i(T) + 2$ ($i = 1, 2$) cannot be reduced any further, which also means that this result is sharp. Additionally, a star illustrates the necessity for an additional point in addition to $V(T)$. The bound $|a - b| \leq 2$ cannot be increased any further. An example is a tree $T(V_1, V_2)$, which is obtained by connecting the centers of two vertex-disjoint stars of order 3 with a path of length 2. Then T is an $(2, 5)$ -bipartite graph and for every $x \in V_1$, $d_T(x) = 3$ (see Fig. 1).

Clearly, there is no 2-packing of T in any B_8 satisfying the requirement of Theorem 1.5. Combining with Theorem 1.3, we obtain a corollary from Theorem 1.5 as follows:

Corollary 1.6. *Let T be a tree with an (a, b) -bipartition such that $|a - b| \leq 2$. Suppose (σ, τ) is a 2-packing of T in the complete bipartite graph $B_{a+b+1}(X_1, X_2)$ such that for any $i \in \{1, 2\}$,*

$$\Delta_i(\sigma(T) \cup \tau(T)) \leq \Delta_i(T) + 2.$$

Then, for a $(|X_1|, |X_2|)$ -bipartite graph G , if

$$(\Delta_1(T) + 2)\Delta_2(G) + \Delta_1(G)(\Delta_2(T) + 2) \leq \max(|X_1|, |X_2|),$$

there exists a packing of two copies of T and G in $B_{a+b+1}(X_1, X_2)$.

In this paper, we will use some special terms and notations. For an (a, b) -bipartite graph $G(U_1, U_2)$, we call vertices of degree equal to 1 in G leaves and vertices of degree greater than 2 in G big. If there is a packing (σ, τ) of G in B_n , we usually regard σ as an identity embedding for G . Without causing confusion, we usually denote $\tau(G)$ by G' . Moreover, for each $u \in V(G)$, we denote the vertex corresponding to G in G' by u' . If $u \neq u'$, then we say that u is 2-placed. If (σ, τ) such that for each $i \in \{1, 2\}$, $\Delta_i(\sigma(G) \cup \tau(G)) \leq \Delta_i(G) + 2$, then we call (σ, τ) a good packing of G in B_n and we often say that there is a good packing of G in B_n .

2. Proof of Theorem 1.5

For a bipartite graph G . If G has a 2-packing in some $B_{|G|+1}$, then we always choose the appropriate 2-packing (θ_1, θ_2) such that every vertex in $B_{|G|+1}$ belongs to at least one of $\theta_1(G)$ and $\theta_2(G)$. Let $T(V_1, V_2)$ be a tree with order n , where $|V_1| = a$, $|V_2| = b$ and $|a - b| \leq 2$. Inspired by the paper [10], we propose the following lemma.

Lemma 2.1 ([10]). *Suppose that for some $i \in \{1, 2\}$, V_i contains a leaf x of T and $xu \in E(T)$ such that $d_T(u) = 2$. If $T(V_1, V_2) - x$ has a good packing in $B_n(X_1, X_2)$, then one of the following statements is true.*

- (a) T has a good packing in some $B_{n+1}(X'_1, X'_2)$, such that for some $i \in \{1, 2\}$, $|X_i| + 1 = |X'_i|$ and $|X_{2-i}| = |X'_{2-i}|$; or
- (b) $|X_i| = 2$.

Table 2
2-packing (σ, τ') of T in B_{n+1} .

	x	t	$v \in V_1 - \{t\}$	$v \in V_2$
σ	x	$\sigma(t)$	$\sigma(v)$	$\sigma(v)$
τ'	z	x	$\tau(v)$	$\tau(v)$

Table 3
2-packing (σ, τ') of T in B_{n+1} .

	y	t	$v \in V'_1 - \{t\}$	x	$v \in V'_2$
σ	y	$\sigma(t)$	$\sigma(v)$	x	$\sigma(v)$
τ'	z	y	$\tau(v)$	x	$\tau(v)$

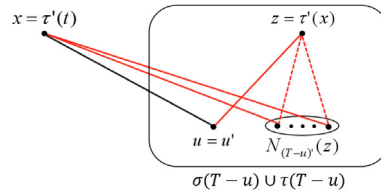


Fig. 2. 2-packing (σ, τ') of T in B_{n+1} .

Proof. W.l.o.g., we may assume that $i = 1$, then $u \in V_2$. By the hypothesis, $T(V_1, V_2) - x$ has a good packing in $B_n(X_1, X_2)$, denoted by (σ, τ) . If $u \neq u'$, we can extend the embeddings σ, τ to T in $B_{n+1}(X_1 \cup \{x\}, X_2)$ by defining $\sigma(x) = \tau(x) = x$. This leads to a good packing of T in $B_{n+1}(X_1 \cup \{x\}, X_2)$. Thus $u = u'$. Suppose for the moment that we have a vertex, denoted by $z \in X_1$, which satisfies $uz \notin T - x$ and $uz \notin (T - x)'$. If z is not in $(T - x)'$, we can expand the embeddings σ, τ to T in $B_{n+1}(X_1 \cup \{x\}, X_2)$ by defining $\sigma(x) = x, \tau(x) = z$. We obtain a good packing of T in $B_{n+1}(X_1 \cup \{x\}, X_2)$. If z belongs to $(T - x)'$, let $t \in T - x$ be such that $\tau(t) = z$. We can define the 2-packing (σ, τ') of T in $B_{n+1}(X_1 \cup \{x\}, X_2)$ as shown in Table 2.

Fig. 2 illustrates the mapping relationships in Table 2, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of $\tau(T)$. It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(X_1 \cup \{x\}, X_2)$. Hence, for each $z \in X_1$, either $uz \in T - u$, or $uz \in (T - u)'$, then

$$2d_T(u) = d_{T-u}(u) + d_{(T-u)'}(u) + 2 = |X_1| + 2.$$

Since $d_T(u) = 2, |X_1| = 2$. This completes the proof of the Lemma. $\square$

We say that $x_0 \dots x_t$ ($t \geq 1$) is a leaf-path of a tree T meaning that $x_0 \dots x_t$ is a path in T where x_0 and x_t are the big and leaf vertices of T , respectively, and for all $i \in \{1, 2, \dots, t-1\}$ ($t \geq 2$), $d_T(x_i) = 2$. We call x_0 the starting point of the leaf path and x_t the ending point of the leaf path. In the following lemmas, we assume that T is a counter-example of Theorem 1.5 of minimum order n . By Theorem 1.4, it is clear that T is not a path or a star. Then the tree T containing leaf-paths of minimal order is either a (2, 3)-bipartite graph or a (3, 3)-bipartite graph or a (2, 4)-bipartite graph.

Lemma 2.2. If T contains leaf-paths, then the length of all leaf-paths is at most 2.

Proof. Suppose that for a contradiction, there exists a leaf-path in T of length at least 3. Let the last two edges of this path be xy and yu , where $d_T(x) = 1$ and $d_T(y) = d_T(u) = 2$. Let $H(V'_1, V'_2) = T - \{x, y\}$, then H is a tree and $\|V'_1\| - \|V'_2\| \leq 2$. W.l.o.g., let $u \in V'_2$. Clearly, $d_H(u) = 1$. It is easy to check that T has a good packing in B_{n+1} when H is a (2, 3)-bipartite graph or a (2, 4)-bipartite graph. Therefore, we assume $|V'_i| \geq 3$ for $i = 1, 2$. By the minimality of T , there is a good packing (σ, τ) of H in some $B_{|H|+1}(U_1, U_2)$. We break into the following two cases.

Case 1. $u = u'$.

Since $d_H(u) = 1$ and $|U_1| \geq |V'_1| \geq 3$, there exists $z \in U_1$ such that $z \in \tau(H)$ and $zu \notin \sigma(H), \tau(H)$. Let $t \in V'_1$ be such that $\tau(t) = z$. We can define the 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{y\}, U_2 \cup \{x\})$ as shown in Table 3.

Fig. 3 illustrates the mapping relationships in Table 3, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of $\tau(H)$. It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(U_1 \cup \{y\}, U_2 \cup \{x\})$.

Case 2. $u \neq u'$.

Since $|U_2| \geq |V'_2| \geq 3$, then there exists $z \in U_2$ such that $z \notin \{u, u'\}$. W.l.o.g., let $z \in \tau(H)$. Then there exists $t \in V'_2$ such that $\tau(t) = z$. We can define the 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{y\}, U_2 \cup \{x\})$ as shown in Table 4.

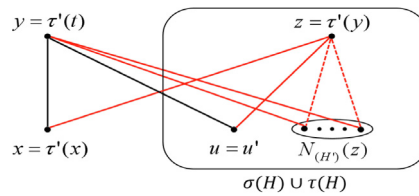


Fig. 3. 2-packing (σ, τ') of T in B_{n+1} .

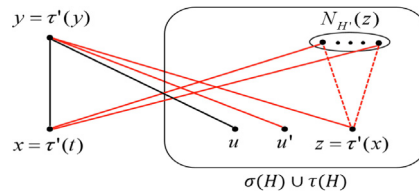


Fig. 4. 2-packing (σ, τ') of T in B_{n+1} .

Table 4
2-packing (σ, τ') of T in B_{n+1} .

	y	$v \in V'_1$	x	t	$v \in V'_2 - \{t\}$
σ	y	$\sigma(v)$	x	$\sigma(t)$	$\sigma(v)$
τ'	y	$\tau(v)$	z	x	$\tau(v)$

Table 5
2-packing (σ, τ') of T in B_{n+1} .

	x_2	$v \in V'_1$	x_1	y_2	$v \in V'_2 - \{x_1\}$
σ	w_2	$\sigma(v)$	x_1	x_2	$\sigma(v)$
τ'	w_1	$\tau(v)$	y_2	x_2	$\tau(v)$

Fig. 4 illustrates the mapping relationships in Table 4, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of H' . It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(U_1 \cup \{y\}, U_2 \cup \{x\})$. This completes the proof of the lemma. $\square$

Lemma 2.3. The starting points of all leaf-paths of length 2 in T are in the same part.

Proof. Suppose that for a contradiction, there exist two leaf-paths $x_0x_1x_2$ and $y_0y_1y_2$ of length 2 with starting points in different parts of T . Here x_0 and y_0 are big vertices. W.l.o.g., let $x_0 \in V_1$ and $y_0 \in V_2$, then $x_2 \in V_1$, $y_2 \in V_2$. Let $H(V'_1, V'_2) = T - \{x_2, y_2\}$, then H is a tree and $\|V'_1\| - \|V'_2\| \leq 2$. Note that at this point $d_H(x_1) = d_H(y_1) = 1$. By the minimality of T , there is a good packing (σ, τ) of H in some $B_{|H|+1}(U_1, U_2)$. Let $A = \{x_1, y_1\}$ and $A' = \{x'_1, y'_1\}$. Clearly we have $0 \leq |A \cap A'| \leq 2$ holds. Let $\{w_1, w_2\} \subset V(B_{|H|+1})$, where $w_1 \notin \tau(H)$, $w_2 \notin \sigma(H)$. It is easy to see that w_1 and w_2 are in the same part. We break into the following three cases.

Case 1. $|A \cap A'| = 0$.

In this case, both x_1 and y_1 are 2-placed. Then we can extend the embeddings σ, τ to the embeddings of T in $B_{n+1}(U_1 \cup \{x_2\}, U_2 \cup \{y_2\})$ by defining $\sigma(y_2) = \tau(y_2) = y_2$, $\sigma(y_1) = \tau(y_1) = y_1$. Clearly (σ, τ) is a good packing of T in B_{n+1} .

Case 2. $|A \cap A'| = 1$.

In this case, we have $x_1 = x'_1$, $y_1 \neq y'_1$ or $x_1 \neq x'_1$, $y_1 = y'_1$ holds. W.l.o.g., we assume that the former holds. If $\{w_1, w_2\} \subseteq U_1$, we can define the 2-packing (σ, τ') of T in $B_{n+1}(U_1, U_2 \cup \{x_2, y_2\})$ as shown in Table 5.

Fig. 5 illustrates the mapping relationships in Table 5, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of H' . It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(U_1, U_2 \cup \{x_2, y_2\})$.

If $\{w_1, w_2\} \subseteq U_2$, we can define the 2-packing (σ', τ') of T in $B_{n+1}(U_1 \cup \{x_2, y_2\}, U_2)$ as shown in Table 6.

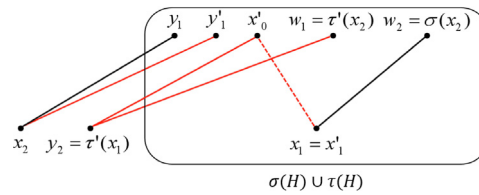
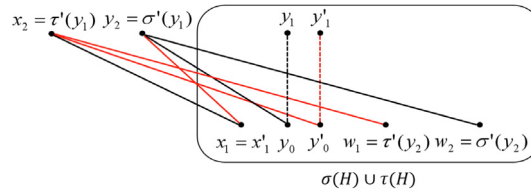
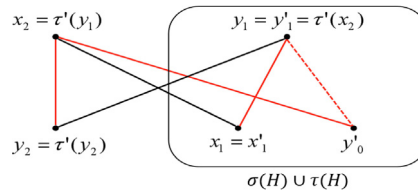
Fig. 5. 2-packing (σ, τ') of T in B_{n+1} .Fig. 6. 2-packing (σ', τ') of T in B_{n+1} .Fig. 7. 2-packing (σ, τ') of T in B_{n+1} .

Table 6
2-packing (σ', τ') of T in B_{n+1} .

	x_2	y_1	$v \in V'_1 - \{y_1\}$	y_2	$v \in V'_2$
σ'	x_2	y_2	$\sigma(v)$	w_2	$\sigma(v)$
τ'	y_2	x_2	$\tau(v)$	w_1	$\tau(v)$

Table 7
2-packing (σ, τ') of T in B_{n+1} .

	x_2	y_1	$v \in V'_1 - \{y_1\}$	y_2	$v \in V'_2$
σ	x_2	y_1	$\sigma(v)$	y_2	$\sigma(v)$
τ'	y_1	x_2	$\tau(v)$	y_2	$\tau(v)$

Fig. 6 illustrates the mapping relationships in Table 6, where the black solid lines indicate the edges of $\sigma'(T)$, the red solid lines indicate the edges of $\tau'(T)$, the black dashed lines indicate the edges of $\sigma(H)$, and the red dashed lines indicate the edges of $\tau(H)$. It is easy to check that (σ', τ') is a good packing for T in $B_{n+1}(U_1 \cup \{x_2, y_2\}, U_2)$.

Case 3. $|A \cap A'| = 2$.

In this case, we can get $x_1 = x'_1$ and $y_1 = y'_1$. Then we can define the 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{x_2\}, U_2 \cup \{y_2\})$ as shown in Table 7.

Fig. 7 illustrates the mapping relationships in Table 7, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of $\tau(H)$. It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(U_1 \cup \{x_2\}, U_2 \cup \{y_2\})$. This proves Lemma 2.3. $\square$

In order to clarify the main proof concepts, we need to add some additional terminology. Let a_1, a_2, a_3 be the number of all leaf vertices, the number of vertices with degree equal to 2, and the number of big vertices in V_1 , respectively. Let b_1, b_2, b_3 be the number of all leaf vertices, the number of vertices with degree equal to 2, and the number of big vertices in V_2 , respectively. Then we have the following claim.

Claim 1. The following three assertions hold.

(a) If $|a - b| = 0$, then

$$a_1 \geq a_3 + 1 \text{ and } b_1 \geq b_3 + 1. \quad (1)$$

(b) If $|a - b| = 1$, then

$$a_1 \geq a_3 + 2 \text{ and } b_1 \geq b_3. \quad (2)$$

(c) If $|a - b| = 2$, then

$$a_1 \geq a_3 + 3 \text{ and } b_1 \geq b_3 - 1. \quad (3)$$

Proof. By the hypothesis, we have $a = a_1 + a_2 + a_3$ and $b = b_1 + b_2 + b_3$. For (a), since $a = b$ and

$$n - 1 = \frac{1}{2} \sum_{v \in V(T)} d_T(v),$$

we have

$$2a - 1 \geq a_1 + 2a_2 + 3a_3 \text{ and } 2b - 1 \geq b_1 + 2b_2 + 3b_3.$$

Substituting $a = a_1 + a_2 + a_3$ and $b = b_1 + b_2 + b_3$ into the above inequality, then (1) holds. Similarly, we can prove (b) and (c). $\square$

Let A_3 and B_3 be the sets consisting of all big vertices in V_1 and V_2 , respectively. If $A_3 \neq \emptyset$, let $A_3 = \{x_1, \dots, x_k\}$, where $k = a_3$. If $B_3 \neq \emptyset$, let $B_3 = \{y_1, \dots, y_l\}$, where $l = b_3$. Clearly A_3 and B_3 are not simultaneously equal to the empty set, otherwise T is a path.

By assertions (a), (b), and (c) of Claim 1, we have $a_1 \geq a_3$. Then we can select k leaves in V_1 and let A_1 denote the set consisting of these leaves. If $b_1 \geq b_3$, we can choose l leaves in V_2 and denote the set consisting of these leaves by B_1 . When A_3 or B_3 is the empty set A_1 or B_1 is also the empty set. If $A_1 \neq \emptyset$, let $A_1 = \{u_1, \dots, u_k\}$, where $k = a_3$. If $B_1 \neq \emptyset$, let $B_1 = \{v_1, \dots, v_l\}$, where $l = b_3$. In this case, let (L, R) be a division of $V(T)$ where

$$L = A_1 \cup A_3 \cup B_1 \cup B_3, \quad R = V(T) - L.$$

We denote by D the set consisting of all leaves of leaf-paths of length 1 in T .

Lemma 2.4. A good packing of T exists in some B_{n+1} if there exists a (L, R) -division of $V(T)$ such that $A_1 \cup B_1 \subseteq D$.

Proof. By assumption, let (L, R) be a division of $V(T)$ such $A_1 \cup B_1 \subseteq D$. We claim that $A_3 \neq \emptyset$ and $B_3 \neq \emptyset$. For a contradiction, w.l.o.g., let $A_3 = \emptyset$. Then $B_3 \neq \emptyset$ (Because A_3 and B_3 are not simultaneously the empty set). By the definition of B_1 , we have $B_1 \neq \emptyset$. Since $B_1 \subseteq D$, then for each $v \in B_1$, v is a leaf in a leaf path of length 1. Thus a neighbor of v is a big vertex, contradicting $A_3 = \emptyset$. Hence the claim is true. In the following we assume that $A_3 \neq \emptyset$ and $B_3 \neq \emptyset$.

If $T[R]$ is a bipartite graph, let (R_1, R_2) be its bipartition. Clearly $e(T[R]) \leq |R| - 1$ since T is a tree. By Theorem 1.4, there is a 2-packing (σ_1, τ_1) of $T[R]$ in $B_{|R|+1}$. Then (σ_1, τ_1) is a good packing of $T[R]$ in $B_{|R|+1}(X_1, X_2)$ since $1 < d_T(x) \leq 2$ for each $x \in R$. Let $\{w_1, w_2\} \subset V(B_{|R|+1})$ such that $w_1 \notin \tau_1(T[R])$ and $w_2 \notin \sigma_1(T[R])$. Then w_1 and w_2 are in the same part.

Let $G_1(A_1, B_3) = T[A_1 \cup B_3]$ and $G_2(A_3, B_1) = T[A_3 \cup B_1]$ (see Fig. 8). Clearly $e(G_1) \leq |A_1|$, $e(G_2) \leq |B_1|$ and hence $e(G_1) + e(G_2) \leq a_1 + b_1$. We begin with by assuming that $\{G_1, G_2\} \notin \Gamma$. By Theorem 1.2, there is a packing of G_1 and G_2 in $B_{a_1+b_1}$. Thus we can pack G_2 into G_1 to get a packing (σ_2^1, τ_2^1) of G_1 and G_2 , and pack G_1 into G_2 to get a packing (σ_2^2, τ_2^2) of G_2 and G_1 , where σ_2^1 and σ_2^2 are identity embeddings. At this point, $\tau_2^1(A_3) = A_1$ and $\tau_2^2(B_3) = B_1$. Let $\sigma_2 = \sigma_2^1 \cup \sigma_2^2$ and $\tau_2 = \tau_2^1 \cup \tau_2^2$. Notice that there are no edges of T between A_1 and B_1 , so (σ_2, τ_2) is a 2-packing of $T[L]$ in $B_{|L|}$ and is a good packing. It is observed that the neighbors of every vertex in R and $A_1 \cup B_1$ are in $A_3 \cup B_3$ and every vertex in $A_3 \cup B_3$ has neighbors in R of degree at most 1, so we can extend (σ_2, τ_2) to the 2-packing of tree T . If $T[R]$ is a bipartite graph, then $(\sigma_1 \cup \sigma_2, \tau_1 \cup \tau_2)$ is a good packing of T in B_{n+1} . If $T[R]$ is not a bipartite graph, we define $\sigma_1(x) = \tau_1(x) = x$ for each $x \in R$ and thus obtain a good packing $(\sigma_1 \cup \sigma_2, \tau_1 \cup \tau_2)$ of T in B_{n+1} .

Therefore assume that $\{G_1, G_2\} \in \Gamma$. W.l.o.g., let G_2 exist a vertex in A_3 , denoted x_1 such that $d_{G_2}(x_1) = b_1$, and for each $u \in A_1$, $d_{G_1}(u) > 0$. Then $d_{G_1}(u) = 1$ for each $u \in A_1$ and $d_{G_2}(v) = 1$ for each $v \in B_1$. Let $u_1 \in A_1$ and $y_1 \in B_3$ such that $u_1 y_1 \in E(G_1)$ and let $v_1 \in B_1$ such that $x_1 v_1 \in E(G_2)$. When $T[R]$ is a bipartite graph, if $\{w_1, w_2\} \subset X_1$, let $G_1^* = G_1 - u_1 y_1$. Then $e(G_1^*) + e(G_2) \leq a_1 + b_1$ and $\{G_1^*, G_2\} \notin \Gamma$. By Theorem 1.2, there is a packing of G_1^* and G_2 in $B_{a_1+b_1}$. Thus, we can get a good packing (σ_2, τ_2) of $T[L] - u_1 y_1$ in $B_{|L|}$ by packing G_2 into G_1^* and packing G_1^* into G_2 as in the above paragraph. Let (L_1, L_2) be the bipartition of $T[L] - u_1 y_1$. Since the neighbors of every vertex in R and $A_1 \cup B_1$ are in $A_3 \cup B_3$, we can define a 2-packing (σ, τ) of T in $B_{n+1}(L_1 \cup X_1, L_2 \cup X_2)$ as shown in Table 8. If $\{w_1, w_2\} \subset X_2$, let $G_2^* = G_2 - x_1 v_1$. Then $e(G_1) + e(G_2^*) \leq a_1 + b_1$ and $\{G_1, G_2^*\} \notin \Gamma$. By Theorem 1.2, there is a packing of G_1 and G_2^* in $B_{a_1+b_1}$. Similarly, we can get a good packing (σ_2, τ_2) of $T[L] - x_1 v_1$ in $B_{|L|}$ by packing G_2^* into G_1 and packing G_1 into G_2^* . Let (L_1, L_2) be the bipartition of $T[L] - x_1 v_1$. We can define the 2-packing (σ, τ) of T in $B_{n+1}(L_1 \cup X_1, L_2 \cup X_2)$ as shown in Table 9. It is easy to check that (σ, τ) is a good packing for T in B_{n+1} .

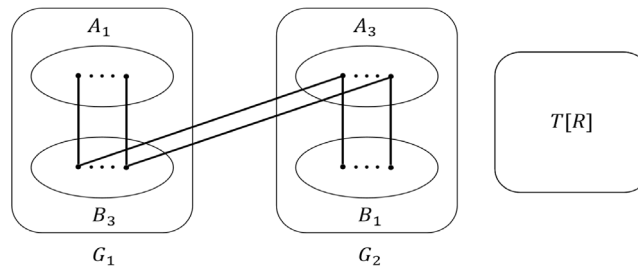


Fig. 8. G_1 , G_2 and $T[R]$.

Table 8

2-packing (σ, τ) of T in B_{n+1} .

	$x \in L_1 - \{u_1\}$	u_1	$x \in R_1$	$x \in L_2$	$x \in R_2$
σ	$\sigma_2(x)$	w_2	$\sigma_1(x)$	$\sigma_2(x)$	$\sigma_1(x)$
τ	$\tau_2(x)$	w_1	$\tau_1(x)$	$\tau_2(x)$	$\sigma_1(x)$

Table 9

2-packing (σ, τ) of T in B_{n+1} .

	$x \in L_1$	$x \in R_1$	$x \in L_2 - \{v_1\}$	v_1	$x \in R_2$
σ	$\sigma_2(x)$	$\sigma_1(x)$	$\sigma_2(x)$	w_2	$\sigma_1(x)$
τ	$\tau_2(x)$	$\tau_1(x)$	$\tau_2(x)$	w_1	$\tau_1(x)$

Table 10

A 2-packing (σ, τ) of $T[L]$ in $B_{|L|+1}$.

	x_1	x_2	$\dots$	x_k	u_1	u_2	$\dots$	u_k	v_1	v_2	$\dots$	v_l	y_1	y_2	$\dots$	y_l
σ	x_1	x_2	$\dots$	x_k	u_1	u_2	$\dots$	u_k	v_1	v_2	$\dots$	v_l	y_1	y_2	$\dots$	y_l
τ	w	u_2	$\dots$	u_k	u_1	x_2	$\dots$	x_k	y_1	y_2	$\dots$	y_l	v_1	v_2	$\dots$	v_l

When $T[R]$ is not a bipartite graph, we can define the 2-packing (σ, τ) of $T[L]$ in $B_{|L|+1}(A_1 \cup A_3 \cup \{w\}, B_1 \cup B_3)$ as shown in Table 10. It is easy to check that (σ, τ) is a good packing for $T[L]$ in $B_{|L|+1}$. At this point, $\tau(A_3) = A_1 \cup \{w\} - \{u_1\}$ and $\tau(B_3) = B_1$. Since the neighbors of every vertex in R and $A_1 \cup B_1$ are in A_3 or B_3 , we can expand the embeddings σ, τ to T in B_{n+1} by defining $\sigma(x) = \tau(x) = x$ for each $x \in R$. Then (σ, τ) is a good packing of T in B_{n+1} . This completes the proof. $\square$

We complete our proof of Theorem 1.5. Let tree $T(V_1, V_2)$ be the counterexample of minimal order n with an (a, b) -bipartition in Theorem 1.5. Lemma 2.2 implies that the length of all leaf-paths of T is at most 2. First, we assume that the conclusion of Theorem 1.5 holds when $|a - b| = 1$. Then, we will prove that the conclusion of Theorem 1.5 holds when $|a - b| = 0$.

Let T be a tree with (a, a) -bipartition. We claim that there is no leaf-path in T of length equal to 2. Assume that for a contradiction, the claim fails. Let $x_0x_1x_2$ be a leaf-path in T of length 2. W.l.o.g., let $x_2 \in V_1$. Let $H(V'_1, V'_2) = T - \{x_2\}$, then $\|V'_1\| - \|V'_2\| = 1$ and hence there is a good packing of H in $B_{|H|+1}$. By Lemma 2.1, we may assume $|X_1| = 2$. Then H is a $(2, 3)$ -bipartite graph and hence T is a $(3, 3)$ -bipartite graph. Obviously we can easily get a good packing of T in B_7 . The claim therefore holds. We assume that all leaf-paths in T are of length 1. By Claim 1, for any partition (L, R) of $V(T)$, the vertices in $A_1 \cup B_1$ are leaves in a leaf-path of length 1. By Lemma 2.4, there is a good packing of T in B_{n+1} .

In the following, we divide the proof into two cases, according to whether $|a - b| = 1$ or $|a - b| = 2$ in Theorem 1.5 in two separate sections.

2.1. The proof of Theorem 1.5 when $|a - b| = 1$

W.l.o.g., let $b = a - 1$, then the tree $T(V_1, V_2)$ is an $(a, a - 1)$ -bipartite graph. Because T contains at least one leaf-path, it follows that $a \geq 3$. If the length of all leaf-paths of T is 1, then $A_1 \cup B_1 \subseteq D$ for any (L, R) -partition of T , which contradicts with Lemma 2.4. Therefore, we can assume that T has at least one leaf-path of length 2.

Lemma 2.5. T contains at most one leaf-path of length 2 with starting point at V_1 .

Proof. Suppose on the contrary that there is two leaf-paths of length 2 in T with starting point at V_1 , say $x_0x_1x_2$ and $y_0y_1y_2$, where x_0 and y_0 are big vertices. Let $H(V'_1, V'_2) = T - \{x_2, y_2\}$. Then H is a tree and $\|V'_1\| - \|V'_2\| = 1$. By the

minimality of T , there is a good packing (σ, τ) of H in some $B_{|H|+1}(U_1, U_2)$. Let $B = \{x_1, y_1\}$ and $B' = \{x'_1, y'_1\}$. Note that $B \cup B' \subset U_2$. Let $V(B_{n+1}(X_1, X_2)) = V(B_{|H|+1}(U_1 \cup \{x_2, y_2\}, U_2))$. For each $c \in \{\sigma, \tau\}$, we add a set E_c of two independent edges between $\{x_2, y_2\}$ and $B \cap B'$ to $c(H)$ and therefore obtain a copy of T in B_{n+1} . Moreover, we choose E_σ and E_τ such that $E_\sigma \cap E_\tau = \emptyset$. Then we extend σ and τ to an embedding of T in B_{n+1} such that (σ, τ) becomes a 2-packing of T in B_{n+1} . Since $d_H(x_1) = d_H(y_1) = 1$, (σ, τ) is a good packing of T in B_{n+1} . This proves Lemma 2.5. $\square$

Observation 2.6. *The starting points of all leaf-paths of length 2 of T are in V_2 and there exists a vertex $v \in B_1$ such that $v \notin D$ for any (L, R) -division of $V(T)$.*

Proof. Lemma 2.3 implies that the starting points of all leaf-paths of length 2 of T are either in V_1 or in V_2 . Assume that the former holds, then by Lemma 2.5 there is at most one such leaf-path. Since there are no leaf-paths of length 2 with starting point in V_2 , there exists $B_1 \subseteq D$ for any (L, R) -division of $V(T)$. By assertion (b) of Claim 1, we have $a_1 \geq a_3 + 2$. Then there exists a leaf set $A_1 \subseteq V_1$ such that every leaf in A_1 is in a leaf-path of length 1. Thus, we can choose a suitable (L, R) -division of T such that $A_1 \cup B_1 \subseteq D$, which contradicts Lemma 2.4. Therefore, the starting points of all leaf-paths of length 2 of T are in V_2 . At this point, $A_1 \subseteq D$ for any (L, R) -division of T . If there exists a (L, R) -division of $V(T)$ such that $B_1 \subseteq D$, then this (L, R) -division also satisfies $A_1 \cup B_1 \subseteq D$. However, this contradicts with Lemma 2.4. Consequently, there exists a vertex $v \in B_1$ such that $v \notin D$ for any (L, R) -division of $V(T)$. $\square$

By Observing 2.6, we assume that the starting points of all leaf-paths of length 2 of T are in V_2 and that for any (L, R) -division of $V(T)$, there exists at least one leaf at B_1 that is a leaf in a leaf-path of length 2 of T . We denote by $N = \{v_1, \dots, v_t\}$ the set consisting of all leaves in the leaf-path of length 2 in B_1 , where $t \leq l$. Let $Z = \{z_1, \dots, z_t\}$ such that $v_i z_i \in E(T)$ for all $i = 1, \dots, t$. Then $d_T(z_i) = 2$ for all $i = 1, \dots, t$. We denote by $M = \{y_1, \dots, y_s\}$ the set of neighbors of Z in B_3 , then $s \leq t$. Let $L^* = L \cup Z$ and $R^* = V(T) - L^*$. We claim that $e(T[R^*]) \leq |R^*| - 2$ because T is a tree and $L^* \neq \emptyset$. The following observation will be useful in the proof of Theorem 1.5.

Observation 2.7. *Given an appropriate (L^*, R^*) -division of $V(T)$. If there is a good packing (σ, τ) of $T[L^*]$ in $B_{|L^*|+1}$ such that $\tau(A_1) = A_3$, $\tau(A_3) = A_1$, $\tau(B_1) = B_3$ and $\tau(B_3) = B_1$, then there is a good packing of T in B_{n+1} .*

Proof. For any (L^*, R^*) -division of tree $V(T)$, we can observe that the neighbors of every vertex in R^* are in $A_3 \cup B_3$ and every vertex in $A_3 \cup B_3$ has neighbors in R of degree at most 1. Assuming that the conditions hold, then we can extend (σ, τ) to be a good packing of T in B_{n+1} . If $T[R^*]$ is not a bipartite graph, we can obtain a good packing of T in B_{n+1} by defining $\sigma(x) = \tau(x) = x$ for each $x \in R^*$.

If $T[R^*]$ is a bipartite graph, let (R_1, R_2) be the bipartition of $T[R^*]$. By assertion (b) of Claim 1, we have $a_1 \geq a_3 + 2$, which implies the presence of at least two leaves in R_1 . This implies that $|R_1| \geq 2$ and that there are at least two isolated vertices of $T[R^*]$ in R_1 . When $|R_2| \geq 2$, it is clear that $T[R^*] \notin \Omega$. Consequently, combine with Theorem 1.1, there is a 2-packing of $T[R^*]$ in $B_{|R^*|}$. When $|R_2| = 1$, $T[R^*]$ consists of the union of a star of order at most 3 and some isolated vertices of at least more than two. In this case, it is clear that there is a 2-packing of $T[R^*]$ in $B_{|R^*|}$. Since the degree of each vertex in $T[R^*]$ is at most 2, the 2-packing of $T[R^*]$ in $B_{|R^*|}$ is also a good packing. Let (σ_1, τ_1) to be a good packing of $T[R^*]$ in $B_{|R^*|}$, then $(\sigma \cup \sigma_1, \tau \cup \tau_1)$ is a good packing of T in B_{n+1} . This completes the proof. $\square$

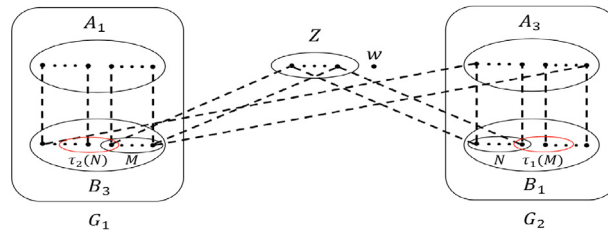
Construction a good packing of $T[L^*]$ in $B_{|L^*|+1}$

We first assume that $A_3 = \emptyset$. Then we have $A_1 = \emptyset$ and $N = B_1$, since the starting point of a leaf path of length 1 must be the big vertex. Let $G_1(X_1, Y_1) = T[Z \cup B_3] \cup \{w\}$, where $X_1 = Z \cup \{w\}$ and $Y_1 = B_3$. Let $G_2(X_2, Y_2) = T[Z \cup B_1] \cup \{w\}$, where $X_2 = Z \cup \{w\}$ and $Y_2 = B_1$. Then both G_1 and G_2 are $(|X_1|, |Y_1|)$ -bipartite graphs. Clearly we have $e(G_1) + e(G_2) = |X_1| - 1 + |Y_1| < |X_1| + |Y_1|$. By Theorem 1.2, there is a packing of G_1 and G_2 in $B_{|X_1|+|Y_1|}$. Then we can get a packing (σ_1, τ_1) of G_1 and G_2 in $B_{|G_2|}$ by packing G_1 into G_2 . Similarly, we can obtain a packing (σ_2, τ_2) of G_1 and G_2 in $B_{|G_1|}$ by packing G_2 into G_1 and such that $\tau_2(x) = \tau_1(x)$ for each $x \in X_1$. Then $(\sigma, \tau) = (\sigma_1 \cup \sigma_2, \tau_1 \cup \tau_2)$ is a good packing of $T[L^*]$ in $B_{|L^*|+1}$ such that $\tau(B_3) = B_1$ and $\tau(B_1) = B_3$, where $V(B_{|L^*|+1}) = L^* \cup \{w\}$.

Now let us assume that $A_3 \neq \emptyset$. Let $G_1 = T[A_1 \cup B_3]$ and $G_2 = T[A_3 \cup B_1]$ (see Fig. 9), then $e(G_1) = k$ and $e(G_2) < l$ since $N \neq \emptyset$. Therefore we have $e(G_1) + e(G_2) < k + l$. By Theorem 1.2, there is a packing of G_1 and G_2 in B_{k+l} . Then we can get a packing (σ_1, τ_1) of G_1 and G_2 in $B_{|G_2|}$ by packing G_1 into G_2 . Similarly, we can obtain a packing (σ_2, τ_2) of G_1 and G_2 in $B_{|G_1|}$ by packing G_2 into G_1 . Let $W_1(X_1, Y_1) = T[Z \cup M] \cup \{w\}$, where $X_1 = Z \cup \{w\}$ and $Y_1 = M$. Let $W_2(X_2, Y_2) = T[Z \cup \tau_1(M)] \cup \{w\}$, where $X_2 = Z \cup \{w\}$ and $Y_2 = \tau_1(M)$. Then $e(W_1) = t$ and $e(W_2) \leq s$. Therefore we have $e(W_1) + e(W_2) < s + t + 1$. By Theorem 1.2, there is a packing of W_1 and W_2 in B_{s+t+1} . Then we can get a packing (σ_3, τ_3) of W_1 and W_2 in $B_{|W_2|}$ by packing W_1 into W_2 . Let $W_3(X_3, Y_3) = T[Z \cup N] \cup \{w\}$, where $X_3 = Z \cup \{w\}$ and $Y_3 = N$. Let $W_4(X_4, Y_4) = T[Z \cup \tau_2(N)] \cup \{w\}$, where $X_4 = Z \cup \{w\}$ and $Y_4 = \tau_2(N)$. Then $e(W_3) = t$ and $e(W_4) \leq t$. Therefore we have $e(W_3) + e(W_4) < t + 1 + t$. By Theorem 1.2, there is a packing of W_3 and W_4 in B_{2t+1} . Then we can get a packing (σ_4, τ_4) of W_3 and W_4 in $B_{|W_4|}$ by packing W_3 into W_4 and such that $\tau_4(x) = \tau_3(x)$ for each $x \in X_1$. Then

$$(\sigma_1 \cup \sigma_2 \cup \sigma_3 \cup \sigma_4, \tau_1 \cup \tau_2 \cup \tau_3 \cup \tau_4)$$

is a good packing of $T[L^*]$ in $B_{|L^*|+1}$, where $V(B_{|L^*|+1}) = L^* \cup \{w\}$. Thus by Observing 2.7, we obtain a good packing of T in B_{n+1} . This completes the proof of Theorem 1.5 when $|a - b| = 1$.

Fig. 9. G_1 , G_2 and $T[L^*]$.

2.2. The proof of Theorem 1.5 when $|a - b| = 2$

W.l.o.g., let $b = a - 2$, then the tree $T(V_1, V_2)$ is an $(a, a - 2)$ -bipartite graph. Clearly, $a \geq 4$ and T must contain a leaf-path. By Lemma 2.3, we assume that all the starting points of the leaf-paths of length 2 in T are in the same part. If T has a leaf-path of length 2, say $x_0x_1x_2$, with the starting point $x_0 \in V_1$, then for the tree $H(V'_1, V'_2) = T - \{x_2\}$, it holds that $\|V'_1\| - \|V'_2\| = 1$. Consequently, there is a good packing of H in some $B_{|H|+1}$. By Lemma 2.1, since $a \geq 4$, there is a good packing of T in some B_{n+1} , which is a contradiction. Therefore, we assume that the starting points of the leaf-paths of length 2 in T are all in V_2 .

By assertion (c) of Claim 1, we have $a_1 \geq a_3 + 3$ and $b_1 \geq b_3 - 1$. When $b_1 \geq b_3$, we can always find l leaves in V_2 to form the set B_1 . Moreover, because all leaf-paths of length 2 in T have their starting points in V_2 , we can find an (L, R) -division (when there are no leaf-paths of length 2 in T) or (L^*, R^*) -division of $V(T)$ (when T has leaf-paths of length 2). At this point, we can use the method employed for the proof when $|a - b| = 1$ to prove Theorem 1.5. Therefore, we assume that $b_1 = b_3 - 1$. The following claim is elementary in proving Theorem 1.5.

Claim 2. Suppose that $b_1 = b_3 - 1$. Then for any $x \in V_2$, we have $d_T(x) \leq 3$.

Proof. Let the number of vertices in V_2 with degree i be denoted as n_i , where $1 \leq n_i \leq \Delta_2 = k$. Assuming $k > 3$, then $b_1 = n_1$, $b_2 = n_2$, and $b_3 = n_3 + n_4 + \dots + n_k$. Since $a = b + 2$ and

$$n - 1 = \frac{1}{2} \sum_{v \in V(T)} d_T(v),$$

then

$$2b + 1 = n_1 + 2n_2 + 3n_3 + \dots + kn_k = b_1 + 2b_2 + 3n_3 + \dots + kn_k.$$

Since $b = b_1 + b_2 + b_3$, $b_1 = b_3 - 1$, and $b_3 = n_3 + n_4 + \dots + n_k$, it follows that

$$0 = n_4 + 2n_5 + \dots + (k - 1)n_k,$$

this implies that for each $i \in \{4, \dots, k\}$,

$$n_i = 0,$$

as required. $\square$

Lemma 2.8. If T contains two leaf-paths of length 2 with starting points in V_2 and they share the same starting point, then there exists a good packing of T in some B_{n+1} .

Proof. Assuming these two leaf-paths are denoted as $z_0x_1x_2$ and $z_0y_1y_2$, then $\{x_1, y_1\} \subseteq V_1$ and $\{z_0, x_2, y_2\} \subseteq V_2$. Let $H(V'_1, V'_2) = T - \{x_1, x_2, y_1, y_2\}$, then H is a tree and $|V'_1| = |V'_2| + 2$. By the minimality of T , there is a good packing (σ, τ) of H in some $B_{|H|+1}(U_1, U_2)$. If $|V'_1| = 3$, then H is a star which admits a $(3, 1)$ -bipartition. It is easy to observe in this case that T has a good packing in some B_9 . Therefore, we assume that $|V'_1| \geq 4$. We break into the following two cases.

Case 1. $\sigma(z_0) \neq \tau(z_0)$.

By Claim 2, we have $d_H(z_0) = 1$. Then we can extend the embeddings σ, τ to the embeddings of T in $B_{n+1}(U_1 \cup \{x_1, y_1\}, U_2 \cup \{x_2, y_2\})$ by defining $\sigma(x_1) = \tau(x_1) = x_1$, $\sigma(y_1) = \tau(y_1) = y_1$, $\sigma(x_2) = \tau(y_2) = x_2$, $\sigma(y_2) = \tau(x_2) = y_2$. Clearly (σ, τ) is a good packing of T in B_{n+1} .

Case 2. $\sigma(z_0) = \tau(z_0)$.

Since $d_H(z_0) = 1$ and $|U_1| \geq |V'_1| \geq 4$, there exists $u_1, u_2 \in U_1$ such that $u_1, u_2 \in \tau(H)$ and $u_1z_0, u_2z_0 \notin E(\sigma(H))$ and $u_1z_0, u_2z_0 \notin E(\tau(H))$. Let $t_1, t_2 \in V'_1$ be such that $\tau(t_1) = u_1$ and $\tau(t_2) = u_2$. We can define the 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{x_1, y_1\}, U_2 \cup \{x_2, y_2\})$, as shown in Table 11.

Table 11

A 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{x_1, y_1\}, U_2 \cup \{x_2, y_2\})$.

	x_1	y_1	t_1	t_2	$x \in V'_1 - \{t_1, t_2\}$	z_0	x_2	y_2	$x \in V'_2 - \{z_0\}$
σ	x_1	y_1	t_1	t_2	$\sigma(x)$	z_0	x_2	y_2	$\sigma(x)$
τ'	u_1	u_2	x_1	y_1	$\tau(x)$	x_2	y_2	z_0	$\tau(x)$

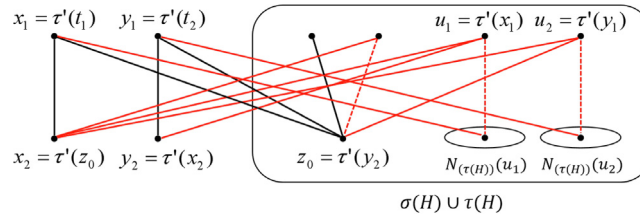


Fig. 10. A 2-packing (σ, τ') of T in $B_{n+1}(U_1 \cup \{x_1, y_1\}, U_2 \cup \{x_2, y_2\})$.

Fig. 10 illustrates the mapping relationships in Table 11, where the black solid lines indicate the edges of $\sigma(T)$, the red solid lines indicate the edges of $\tau'(T)$, and the red dashed lines indicate the edges of $\tau(H)$. It is easy to check that (σ, τ') is a good packing for T in $B_{n+1}(U_1 \cup \{x_1, y_1\}, U_2 \cup \{x_2, y_2\})$. This completes the proof of the lemma. $\square$

Now we will add a vertex w to $V(T)$ and define an (L, R) -partition of $V(T) \cup \{w\}$. Since $a_1 \geq a_3 + 3$, there are at least three leaves in V_1 . Because T does not have leaf-paths of length 2 starting points from V_1 , the neighbors of the leaves in V_1 are all in V_2 . By Claim 2, each vertex in V_2 has a degree of at most 3, so there are at least two big vertices in V_2 (There are at most two leaves with a common starting point in V_1). Recall that A_3 and B_3 represent the number of all big vertices in V_1 and V_2 respectively, with $|A_3| = k$ and $|B_3| = l$. Therefore, it follows that $l \geq 2$. Since $b_1 = b_3 - 1$, there is at least one leaf in V_2 . We can select $l - 1$ leaves in V_2 and let B_1 denote the set consisting of these leaves. We select k leaves in V_1 and let A_1 denote the set consisting of these leaves. When $k = 0$, $A_3 = \emptyset$. If T contains a leaf-path of length 2, then its starting point is in V_2 . We denote by $N = \{v_1, \dots, v_t\}$ the set consisting of all leaves in the leaf-path of length 2 in B_1 , where $t \leq l$. Let $Z = \{z_1, \dots, z_t\}$ such that $v_i z_i \in E(T)$ for all $i = 1, \dots, t$. Then $d_T(z_i) = 2$ for all $i = 1, \dots, t$. By Lemma 2.8, each leaf-path of length 2 in T has a different starting point, so we use $M = \{y_1, \dots, y_t\}$ to denote the set of neighbors of Z in B_3 .

In the following, we define an appropriate (L, R) -division of $V(T) \cup \{w\}$. Suppose that there is no leaf-path of length 2 in T , as there is at least one leaf in V_2 whose neighbor is in V_1 , so $A_3 \neq \emptyset$. Let $A_3^* = A_3$, $A_1^* = A_1$, $B_3^* = B_3$, $B_1^* = B_1 \cup \{w\}$. When T contains only one leaf-path of length 2, denote this leaf-path as $y_1 z_1 v_1$. Let $A_3^* = A_3 \cup \{z_1\}$. Since $a_1 \geq a_3 + 3$, we select a leaf u_0 that is not in A_1 , and let $A_1^* = A_1 \cup \{u_0\}$, $B_3^* = B_3$ and $B_1^* = B_1 \cup \{w\}$. In both cases, we define an (L, R) -division of $V(T) \cup \{w\}$ as

$$L = A_1^* \cup A_3^* \cup B_1^* \cup B_3^*, \quad R = V(T) - (L - \{w\}).$$

When T has at least two leaf-paths of length 2, let $A_3^* = A_3$, $A_1^* = A_1$, $B_3^* = B_3$, and $B_1^* = B_1 \cup \{w\}$. In this situation, we define a (L, R) -division of $V(T) \cup \{w\}$ as

$$L = A_1^* \cup A_3^* \cup B_1^* \cup B_3^* \cup Z, \quad R = V(T) - (L - \{w\}).$$

Let $T_L(L_1, L_2) = T[L - \{w\}] \cup \{w\}$, where $w \in L_2$. Clearly, T_L is a bipartite graph and w is an isolated vertex of T_L .

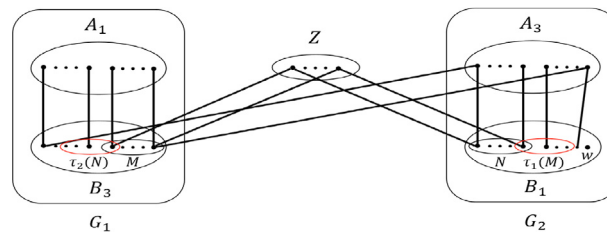
Observation 2.9. Given an appropriate (L, R) -division of $V(T) \cup \{w\}$. If there is a good packing (σ, τ) of T_L in $B_{|L|}$ such that $\tau(A_1^*) = A_3^*$, $\tau(A_3^*) = A_1^*$, $\tau(B_1^*) = B_3^*$ and $\tau(B_3^*) = B_1^*$, then there is a good packing of T in B_{n+1} .

Proof. For any (L, R) -division of $V(T) \cup \{w\}$, we can observe that the neighbors of every vertex in R are in $A_3^* \cup B_3^*$ and every vertex in $A_3^* \cup B_3^*$ has neighbors in R of degree at most 1. If $T[R]$ is not a bipartite graph, we can obtain a good packing of T in B_{n+1} by defining $\sigma(x) = \tau(x) = x$ for each $x \in R$. Clearly, (σ, τ) is a good packing of T in B_{n+1} .

If $T[R]$ is a bipartite graph, let (R_1, R_2) be the bipartition of $T[R]$. Because $a_1 \geq a_3 + 3$ and $|A_1^*| \leq a_1 + 1$, there are at least two leaves of T in R_1 . This implies that $|R_1| \geq 2$ and that there are at least two isolated vertices of $T[R]$ in R_1 . When $|R_2| \geq 2$, then $T[R] \notin \Omega$. By Theorem 1.1, there is a 2-packing of $T[R]$ in $B_{|R|}$. When $|R_2| = 1$, $T[R]$ consists of the union of a star of order at most 3 and some isolated vertices of at least more than two. In this case, it is clear that there is a 2-packing of $T[R]$ in $B_{|R|}$. Since the degree of each vertex in $T[R]$ is at most 2, the 2-packing of $T[R]$ in $B_{|R|}$ is also a good packing. Let (σ_1, τ_1) to be a good packing of $T[R]$ in $B_{|R|}$, then $(\sigma \cup \sigma_1, \tau \cup \tau_1)$ is a good packing of T in B_{n+1} . $\square$

Construction a good packing of T_L in $B_{|L|}$

First, we assume that T has at most one leaf-path of length 2. Let $G_1(A_1^*, B_3^*) = T[A_1^* \cup B_3^*]$. Let $G_2(A_3^*, B_1^*) = T[A_3^* \cup B_1^*] \cup \{w\}$. Then w is an isolated vertex of G_2 . Clearly $e(G_1) \leq |A_1^*|$, $e(G_2) \leq |B_1^*| - 1$ and hence $e(G_1) + e(G_2) < |A_3^*| + |B_1^*|$.

Fig. 11. G_1 , G_2 and T_L .

By Theorem 1.2, there is a packing of G_1 and G_2 in $B_{|A_3^*|+|B_1^*|}$. Thus we can pack G_1 into G_2 to get a packing (σ_2^1, τ_2^1) of G_1 and G_2 , and pack G_2 into G_1 to get a packing (σ_2^2, τ_2^2) of G_2 and G_1 , where σ_2^1 and σ_2^2 are identity embeddings. At this point, $\tau_2^1(B_3^*) = B_1^*$ and $\tau_2^2(A_3^*) = A_1^*$. Let $\sigma = \sigma_2^1 \cup \sigma_2^2$ and $\tau = \tau_2^1 \cup \tau_2^2$. Notice that there are no edges of T between A_1^* and B_1^* , so (σ, τ) is a 2-packing of T_L in $B_{|L|}$ and is a good packing. At this point, $\tau(A_1^*) = A_3^*$, $\tau(A_3^*) = A_1^*$, $\tau(B_1^*) = B_3^*$ and $\tau(B_3^*) = B_1^*$.

Now, we assume that T has at least two leaf-paths of length 2, then $|Z| \geq 2$. Suppose that $A_3^* = \emptyset$. Then $A_1 = \emptyset$ and $N = B_1$. Let $G_1(X_1, Y_1) = T[Z \cup B_3^*]$, where $X_1 = Z$ and $Y_1 = B_3^*$. Let $G_2(X_2, Y_2) = T[Z \cup B_1] \cup \{w\}$, where $X_2 = Z$ and $Y_2 = B_1 \cup \{w\} = B_1^*$. Then both G_1 and G_2 are $(|X_1|, |Y_1|)$ -bipartite graphs. Clearly we have $e(G_1) + e(G_2) = |X_1| + |Y_1| - 1 < |X_1| + |Y_1|$. By Theorem 1.2, there is a packing of G_1 and G_2 in $B_{|X_1|+|Y_1|}$. Then we can get a packing (σ_1, τ_1) of G_1 and G_2 in $B_{|G_2|}$ by packing G_1 into G_2 . Similarly, we can obtain a packing (σ_2, τ_2) of G_1 and G_2 in $B_{|G_1|}$ by packing G_2 into G_1 and such that $\tau_2(x) = \tau_1(x)$ for each $x \in X_1$. Then $(\sigma, \tau) = (\sigma_1 \cup \sigma_2, \tau_1 \cup \tau_2)$ is a good packing of T_L in $B_{|L|}$ such that $\tau(B_1^*) = B_3^*$ and $\tau(B_3^*) = B_1^*$.

Therefore, $A_3^* \neq \emptyset$. Let $G_1(A_1^*, B_3^*) = T[A_1^* \cup B_3^*]$ and $G_2(A_3^*, B_1^*) = T[A_3^* \cup B_1^*] \cup \{w\}$ (see Fig. 11). Then $e(G_1) = k$ and $e(G_2) = l - 1$. Therefore we have $e(G_1) + e(G_2) < k + l$. By Theorem 1.2, there is a packing of G_1 and G_2 in B_{k+l} . Then we can get a packing (σ_1, τ_1) of G_1 and G_2 in $B_{|G_2|}$ by packing G_1 into G_2 . Similarly, we can obtain a packing (σ_2, τ_2) of G_1 and G_2 in $B_{|G_1|}$ by packing G_2 into G_1 . Let $W_1(Z, M) = T[Z \cup M]$ and $W_2(Z, \tau_1(M)) = T[Z \cup \tau_1(M)]$. Then $e(W_1) = t$ and $e(W_2) \leq t$. Therefore we have $e(W_1) + e(W_2) \leq 2t$. By Lemma 2.8 and $|Z| \geq 2$, we can deduce that W_1 is a matching. Each component of W_2 is either K_2 or an isolated vertex. Therefore, $\{W_1, W_2\} \notin \Gamma$. By Theorem 1.2, there is a packing of W_1 and W_2 in B_{2t} . Then we can get a packing (σ_3, τ_3) of W_1 and W_2 in $B_{|W_2|}$ by packing W_1 into W_2 . Let $W_3(Z, N) = T[Z \cup N]$ and $W_4(Z, \tau_2(N)) = T[Z \cup \tau_2(N)]$. Then $e(W_3) = t$ and $e(W_4) \leq t$. So we obtain $e(W_3) + e(W_4) \leq 2t$. Clearly, W_3 is a matching. From Lemma 2.8 and $|Z| \geq 2$, we can deduce that each component of W_4 is either K_2 or an isolated vertex. By Theorem 1.2 once again, there is a packing of W_3 and W_4 in B_{2t} . Then we can get a packing (σ_4, τ_4) of W_3 and W_4 in $B_{|W_4|}$ by packing W_3 into W_4 and such that $\tau_4(x) = \tau_3(x)$ for each $x \in Z$. Then

$$(\sigma_1 \cup \sigma_2 \cup \sigma_3 \cup \sigma_4, \tau_1 \cup \tau_2 \cup \tau_3 \cup \tau_4)$$

is a good packing of T_L in $B_{|L|}$. At this point, $\tau(A_1^*) = A_3^*$, $\tau(A_3^*) = A_1^*$, $\tau(B_1^*) = B_3^*$ and $\tau(B_3^*) = B_1^*$. By Observation 2.9, we get a good packing of T in B_{n+1} . This completes the proof of Theorem 1.5.

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Data availability

No data was used for the research described in the article.

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